

LOT® COMPUTER

COSMO® Cube v1.0 – Assembly & QC Manual

CLASSIFICATION: RESTRICTED // S-2 EYES · PRODUCTION-FACING

SCOPE

This manual covers final assembly and QC for the 100-unit pilot run. PCB fabrication and SMT placement are performed by PCBWay per docs/corporate/LOT-COSMO-COMPUTER-BOM.md Section 02; this manual begins at receipt of populated boards.

ENCLOSURE ASSEMBLY

1. Seat the populated PCB into the front shell (polished face), display and camera windows aligned to their cutouts.
2. Place the 150mAh LiPo cell in its pocket, connector routed clear of the wireless charge coil.
3. Align the wireless charge coil into the rear shell (matte face) pocket.
4. Mate front and rear shells. Install 4 × M1.4 recessed screws into the threaded standoffs.
5. Torque to 0.05 N·m. Do not overtorque – stainless standoffs at this size strip above 0.08 N·m.

100-UNIT QC CHECKLIST

#	CHECK	PASS CRITERIA
1	Battery voltage at receipt	3.7V nominal, 3.4-4.2V acceptable range
2	Enclosure seam gap	No visible gap, screws fully seated
3	Wireless charge handshake	Device wakes and reports charging within 5s of pad contact
4	Wi-Fi + SSE handshake	Device reaches SSE STANDBY state (firmware doc Sec. 3) within 30s of boot
5	Pairing flow	SoftAP visible, completes test pairing to a staging LOT profile
6	Display render	Test string renders legibly, no dead pixels/segments
7	Camera capture	One test frame captured and received server-side on COPY press
8	Weather sensor read	Temperature/humidity reading within 2 degrees C / 5% RH of reference
9	Button continuity + debounce	Single press registers exactly one COPY event
10	Laser serial legible + logged	Serial matches hardware_devices record (software doc Sec. 6)

DISPOSITION

A unit failing any check above is set aside, not reworked into the shipping batch without a second full pass of all 10 checks. Pass rate below 95/100 halts the run and reopens LOT-COSMO-COMPUTER-BOM.md Section 01 for a component-level review before continuing.